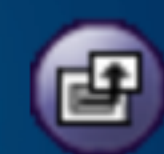


1. A 42-year-old man has had the gradual onset of greasy, diffuse scaling of the scalp with variable itching over the past year. Yellowish red scaling papules appear along the hairline and behind the ears; there is no hair loss. Which of the following is the most likely diagnosis?

- ☐ A) Exfoliative dermatitis
- ☐ B) Lichen simplex chronicus
- ☐ C) Pediculosis capitis
- ☐ D) Psoriasis
- ☐ E) Seborrheic dermatitis



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2. A 24-year-old man comes to the physician for a follow-up examination. Two months ago, he was seen for a red, scaly rash over his groin. A 6-week course of topical clotrimazole resolved the rash completely, but it recurred 2 days ago. He is sexually active with one partner and uses condoms consistently. Examination shows an erythematous, excoriated rash over the groin and a similar rash over the instep of the left foot. A KOH preparation of a skin scraping shows hyphae. Which of the following is the most likely cause of this patient's recurrent infection?

- ☐ A) Autoinfection
- ☐ B) Clotrimazole resistance
- ☐ C) Impaired cellular immunity
- ☐ D) Impaired humoral immunity
- ☐ E) Reinfection from a sexual partner



Previous



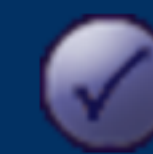
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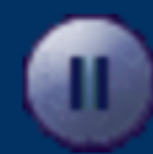
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3. An otherwise healthy 30-year-old man has had recurrent aching right shoulder pain initiated by reaching overhead; it also occurs at night in bed. The pain is most prominent in the area of the deltoid muscle. It is elicited by abduction of the shoulder against resistance. Which of the following is the most likely site of the underlying condition?

- ☐ A) Biceps tendon
- ☐ B) Cervical nerve root
- ☐ C) Deltoid muscle
- ☐ D) Proximal humerus
- ☐ E) Supraspinatus tendon



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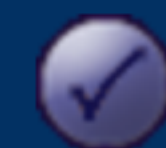
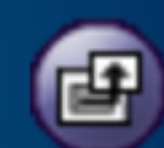
Help



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4. A 72-year-old woman comes to the physician because she is concerned about her high blood pressure. Two days ago, her blood pressure was 200/105 mm Hg at the drug store. She has a 20-year history of hypertension that had been well controlled with a thiazide diuretic. At her last visit 3 months ago, her blood pressure was 140/75 mm Hg. Today, her pulse is 80/min, respirations are 12/min, and blood pressure is 210/114 mm Hg. Funduscopic examination shows arteriovenous nicking. A right carotid bruit is heard. Laboratory studies are within normal limits. Which of the following is the most likely cause of these findings?

- ☐ A) Acute glomerulonephritis
- ☐ B) Acute porphyria
- ☐ C) Coarctation of the aorta
- ☐ D) Cushing syndrome
- ☐ E) Hyperaldosteronism
- ☐ F) Hyperparathyroidism
- ☐ G) Hyperthyroidism
- ☐ H) Pheochromocytoma
- ☐ I) Renal artery stenosis



5. A 57-year-old woman with stage IV non-small cell lung carcinoma is admitted to the hospital because of a 2-day history of shortness of breath and fever. Her temperature is 39°C (102.2°F). She receives the diagnosis of postobstructive pneumonia. Treatment with intravenous antibiotics is begun, and her symptoms resolve 3 days later. During a family meeting to discuss her treatment options, the patient states that she wishes to discontinue chemotherapy and requests no resuscitation if she has a cardiac arrest. In addition to obtaining a signed advance directive and durable power of attorney for health care decisions, which of the following is the most appropriate recommendation for this patient?
- ☐ A) Discharge home with home hospice care
 - ☐ B) Discharge home with physical and occupational therapy
 - ☐ C) Discharge home with visiting nurse services
 - ☐ D) Discharge to a skilled nursing care facility
 - ☐ E) Remain in the hospital



6. An 87-year-old resident of a skilled nursing care facility is brought to the physician by staff because of a lesion on his left leg for 3 weeks. He has been confined to bed for 3 years because of a left hemiplegia following a cerebral infarction. Examination shows a 5 x 7-cm, grade 3 pressure ulcer over the left greater trochanter; there is an overlying eschar with a small rim of erythema. There are no other areas of ulceration. Which of the following is the most appropriate next step in management?

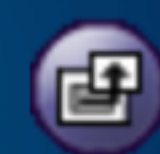
- ☐ A) Use of an air bed
- ☐ B) Application of wet-to-dry dressings
- ☐ C) Application of silver sulfadiazine cream and a sterile dressing
- ☐ D) Debridement of the ulcer
- ☐ E) Skin graft



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7. A 42-year-old woman comes to the physician for a follow-up examination. She was diagnosed with hypertension 6 months ago, and hydrochlorothiazide therapy was begun. Eight weeks ago, her blood pressure was 176/96 mm Hg, and lisinopril was added to her regimen. She says she feels well. She has no other history of serious illness. She occasionally takes ibuprofen for muscle pain or headache. She walks for 30 minutes three times weekly. She is 170 cm (5 ft 7 in) tall and weighs 66 kg (145 lb); BMI is 23 kg/m². Her pulse is 76/min, and blood pressure is 160/98 mm Hg in the right upper extremity and 162/96 mm Hg in the left upper extremity while sitting and standing. Cardiopulmonary examination shows no abnormalities. There is no peripheral edema. Fasting laboratory studies show:

Serum	
Na ⁺	140 mEq/L
K ⁺	4 mEq/L
Cl ⁻	105 mEq/L
HCO ₃ ⁻	24 mEq/L
Urea nitrogen	16 mg/dL
Glucose	94 mg/dL
Creatinine	1 mg/dL
Urine protein	1+

The patient asks how she can decrease her risk for renal failure. Which of the following is the most appropriate recommendation?

- ☐ A) Protein-restricted diet
- ☐ B) 24-Hour urine collection for measurement of protein concentration
- ☐ C) Increasing the dose of lisinopril
- ☐ D) Switching from hydrochlorothiazide to furosemide
- ☐ E) No further measures are indicated at this time

8. A 55-year-old man is admitted to the hospital in a coma following a sudden collapse. He has a history of recent coronary bypass operation for coronary artery disease. ECG shows wide QRS complexes (QRS 160 ms) with a ventricular rate of 170/min and an atrial rate of 110/min. Which of the following is the most appropriate initial step in management?

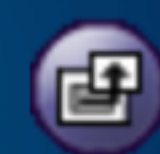
- ☐ A) Administration of digitalis
- ☐ B) Administration of verapamil
- ☐ C) Carotid sinus massage
- ☐ D) Direct current countershock
- ☐ E) External pacing



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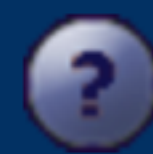
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9. A 32-year-old woman is admitted to the hospital 8 hours after the onset of nausea and vomiting and epigastric abdominal pain that radiates to her back. She has had two similar episodes during the past 2 months. She has no other history of serious illness. She does not drink alcohol. Abdominal examination shows moderate epigastric tenderness. Laboratory studies show:

Hemoglobin	12.5 g/dL
Hematocrit	38%
Leukocyte count	15,400/mm ³
Platelet count	326,000/mm ³
Serum amylase	1450 U/L

Ultrasonography of the gallbladder shows no abnormalities. A CT scan of the abdomen is shown. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) HIDA scan
- ☐ B) Endoscopic retrograde cholangiopancreatography
- ☐ C) Esophagogastroduodenoscopy
- ☐ D) Mesenteric angiography
- ☐ E) Percutaneous transhepatic cholangiography



10. A 27-year-old man comes to the physician because of fatigue and loose stools over the past 3 months; during this period, he has had a 6.8-kg (15-lb) weight loss despite a good appetite and normal dietary intake. He has a childhood history of diarrhea and difficulty gaining weight that resolved without treatment during early adolescence. He has not traveled recently. He does not appear acutely ill. He is 173 cm (5 ft 8 in) tall and weighs 57 kg (125 lb); BMI is 19 kg/m². Vital signs are within normal limits. Examination shows no abnormalities. Test of the stool shows no leukocytes, and cultures of the stool are negative for pathogens. A 72-hour test of the stool for fecal fat shows 12 g/d (N=2–6). D-xylose absorption test shows a serum D-xylose concentration of 4 mg/dL (N=25–40) 2 hours after the administration of 25 g of D-xylose. Which of the following is the most likely underlying cause of this patient's malabsorption?

- ☐ A) Bacterial overgrowth in the small bowel
- ☐ B) Exocrine pancreatic insufficiency
- ☐ C) Failure of bile acid excretion
- ☐ D) Failure of bile acid reabsorption
- ☐ E) Lactase deficiency
- ☐ F) Villous atrophy in the small bowel



11. A 77-year-old woman comes to the physician because of palpitations for 2 days. She has had a 4.5-kg (10-lb) weight loss during the past 6 months. Her temperature is 37.4°C (99.3°F), pulse is 120/min and irregular, respirations are 24/min, and blood pressure is 130/70 mm Hg. Examination shows nontender enlargement of the left lobe of the thyroid gland. Serum studies show a thyroid-stimulating hormone concentration less than 0.05 $\mu\text{U/mL}$ and thyroxine (T_4) concentration of 30 $\mu\text{g/dL}$. A ^{131}I study shows a single area of increased uptake in the left lobe of the thyroid gland and suppressed uptake elsewhere in the gland. Which of the following is the most likely mechanism of these findings?

- ☐ A) Autoimmune destruction of thyroid cells
- ☐ B) Focal granulomatous inflammation
- ☐ C) Parafollicular cell cytokine overproduction
- ☐ D) Production of thyroid-stimulating immunoglobulin
- ☐ E) Autonomous T_4 production



12. A 72-year-old nursing home resident is brought to the emergency department because of abdominal pain and bloody stools for 3 days. She has dementia, Alzheimer type, and a long-standing history of constipation treated with psyllium and docusate. She is in no apparent distress. Her temperature is 39°C (102.2°F), pulse is 88/min, respirations are 18/min, and blood pressure is 150/80 mm Hg. Abdominal examination shows left lower quadrant tenderness and distention. Rectal examination shows blood and no masses. She is not oriented to place or time. Which of the following is the most likely diagnosis?

- | | |
|---|---|
| ☐ A) Campylobacteriosis | ☐ G) Irritable bowel syndrome |
| ☐ B) Celiac disease | ☐ H) Ischemic colitis |
| ☐ C) <i>Clostridium difficile</i> colitis | ☐ I) Salmonellosis |
| ☐ D) Colon cancer | ☐ J) Shigellosis |
| ☐ E) Diverticulitis | ☐ K) Ulcerative colitis |
| ☐ F) Diverticulosis | |



13. A 27-year-old man is brought to the emergency department 40 minutes after the sudden onset of shortness of breath while scuba diving. He reports that he had to ascend rapidly from a 70-foot depth because he was running out of compressed air. During the ascent, he had the sudden onset of midchest pain; he had shortness of breath by the time he reached the surface. He has no history of serious illness and takes no medications. He is in moderate respiratory distress and unable to speak in complete sentences. His respirations are 28/min. Pulse oximetry on room air shows an oxygen saturation of 88%. He is using accessory muscles of respiration. The trachea is midline. Pulmonary examination shows decreased breath sounds and hyperresonance to percussion on the left. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Measurement of serum D-dimer concentration
- ☐ B) Chest x-ray
- ☐ C) ECG
- ☐ D) Echocardiography
- ☐ E) CT scan of the thorax



14. A 62-year-old man is brought to the emergency department 45 minutes after the sudden onset of shortness of breath. When paramedics arrived at his home, he was agitated but alert and oriented to person, place, and time. He was cyanotic and in severe respiratory distress. He was treated with 100% oxygen by face mask en route to the hospital. He has a 15-year history of chronic obstructive pulmonary disease treated with home oxygen therapy (1.5 L/min) over the past 2 years. He has smoked two packs of cigarettes daily for 40 years. On arrival, he is drowsy and awakens only to loud voices or painful stimuli. His temperature is 37°C (98.6°F), pulse is 104/min, respirations are 14/min and shallow, and blood pressure is 164/90 mm Hg. Breath sounds are diffusely decreased, and air movement is poor with prolonged expiration. Which of the following is the most likely underlying cause of this patient's decreased level of consciousness?

- ☐ A) Hypercarbia
- ☐ B) Hypoxemia
- ☐ C) Metabolic acidosis
- ☐ D) Oxygen toxicity
- ☐ E) Respiratory alkalosis



15. A 47-year-old woman with type 2 diabetes mellitus comes to the physician for a follow-up examination. Six weeks ago, her fasting serum glucose concentration was 177 mg/dL; urine protein was 1+. She has been taking metformin and following a low-carbohydrate diet for the past 4 weeks. She has no other medical problems and is taking no other medications. Her father died of end-stage renal disease secondary to type 2 diabetes mellitus. She is 165 cm (5 ft 5 in) tall and weighs 82 kg (180 lb); BMI is 30 kg/m². Her weight is unchanged from her last visit. Today, her blood pressure is 130/82 mm Hg; at her last examination 4 weeks ago, it was 128/80 mm Hg. The remainder of the physical examination shows no abnormalities. Laboratory studies show:

Hemoglobin A _{1c}	7.3%
Serum	
Urea nitrogen	17 mg/dL
Glucose	169 mg/dL
Creatinine	1.1 mg/dL
Urine protein	180 mg/24 h

Which of the following measures is most likely to slow the progression of diabetic nephropathy in this patient?

- ☐ A) Calorie-reduced diet
- ☐ B) Dietary protein restriction to 0.6 g/kg daily
- ☐ C) Maintenance of a hemoglobin A_{1c} of less than 6%
- ☐ D) Addition of lisinopril to the medication regimen
- ☐ E) Switching from metformin to insulin therapy

16. A 20-year-old woman comes for a routine health maintenance examination. Her temperature is 37°C (98.6°F), pulse is 80/min, respirations are 12/min, and blood pressure is 110/70 mm Hg. The S₂ is split and fixed. A grade 2/6 systolic murmur is heard at the third intercostal space. Examination shows no other abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Aortic insufficiency
- ☐ B) Atrial septal defect
- ☐ C) Mitral insufficiency
- ☐ D) Patent ductus arteriosus
- ☐ E) Pulmonic insufficiency
- ☐ F) Tetralogy of Fallot
- ☐ G) Tricuspid insufficiency
- ☐ H) Ventricular septal defect



17. A 52-year-old woman with hypertension comes to the emergency department 45 minutes after the onset of tearing pain that starts at the base of her neck and radiates to her back. On arrival, her temperature is 37°C (98.6°F), pulse is 96/min, respirations are 20/min, and blood pressure is 190/110 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 99%. No other abnormalities are noted on physical examination. An ECG shows sinus tachycardia and left ventricular hypertrophy. A chest x-ray shows a widened mediastinum. The most appropriate next step in management is administration of which of the following?

- ☐ A) Labetalol
- ☐ B) Morphine
- ☐ C) Nitroglycerin
- ☐ D) Phenylephrine
- ☐ E) Sodium nitroprusside



18. A 62-year-old man has had lethargy, weakness, confusion, and constipation for 3 days. Three weeks ago, he was diagnosed with inoperable squamous cell carcinoma of the lung. Examination shows decreased skin turgor. Breath sounds are decreased at the left apex. Laboratory studies show:

Leukocyte count	8100/mm ³
Serum	
Na ⁺	142 mEq/L
K ⁺	4.8 mEq/L
Ca ²⁺	15.2 mg/dL
Phosphorus	1.2 mg/dL
Albumin	4.5 g/dL

An ECG shows a short QT interval. Intravenous administration of 0.9% saline(200 mL/h) is begun. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Allopurinol
- ☐ B) Cisplatin
- ☐ C) Cryoprecipitate
- ☐ D) Demeclocycline
- ☐ E) Doxycycline
- ☐ F) Fresh frozen plasma
- ☐ G) Heparin
- ☐ H) Hydroxyurea
- ☐ I) Pamidronate
- ☐ J) Tamoxifen

19. A 52-year-old man comes to the physician because of fatigue for 4 months. He has hypertension and a 2-year history of chronic renal insufficiency. His serum creatinine concentrations have ranged between 2.5 mg/dL and 3.5 mg/dL. Medications include lisinopril and amlodipine. His pulse is 96/min, and blood pressure is 160/72 mm Hg. Examination shows no other abnormalities except for mild pallor. Test of the stool for occult blood is negative. Laboratory studies show:

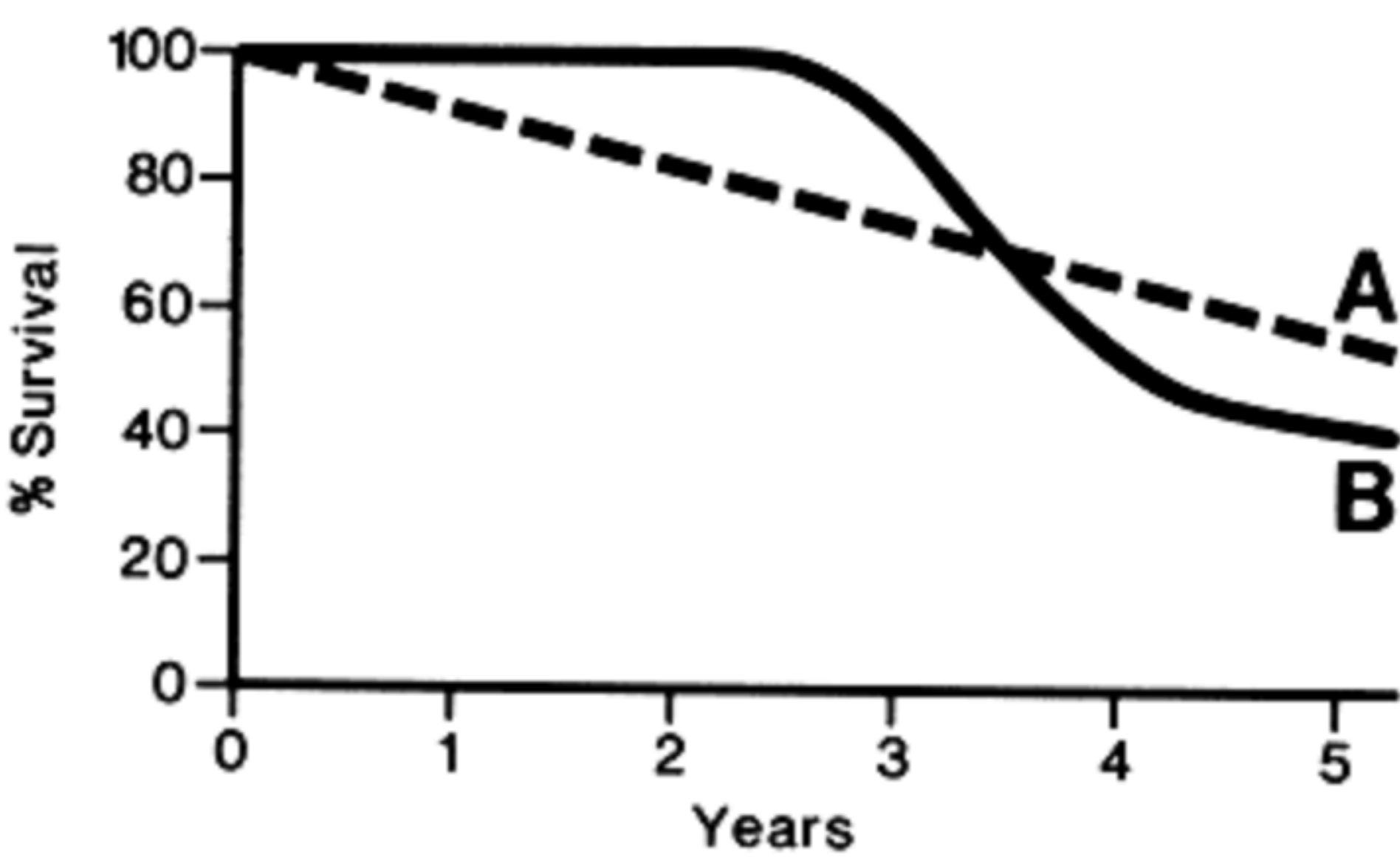
Hematocrit	22%
Mean corpuscular volume	89 μm^3
Reticulocyte count	2.5%
Serum	
Creatinine	3.4 mg/dL
Ferritin	150 ng/mL
Iron	35 $\mu\text{g/dL}$
Transferrin saturation	28% (N=22%–46%)

Supplementation with which of the following is most likely to have prevented these hematologic findings?

- ☐ A) Erythropoietin
- ☐ B) Folic acid
- ☐ C) Iron
- ☐ D) Vitamin B₁₂ (cyanocobalamin)
- ☐ E) No preventive measures are available

20. The life table shown compares the natural disease history of groups A and B. Which group at which time has the poorest 1-year survival rate?

- ☐ A) Group A at Time 0–1
- ☐ B) Group B at Time 0–1
- ☐ C) Group A at Year 2–3
- ☐ D) Group B at Year 3–4
- ☐ E) Group B at Year 4–5



21. A 67-year-old man had the acute onset of bilateral weakness of the lower extremities and urinary incontinence 2 days ago. He has had carcinoma of the prostate for 2 years and bone metastases to the pelvis for 6 months. His temperature is 37.4°C (99.3°F), pulse is 86/min, respirations are 18/min, and blood pressure is 160/86 mm Hg. There is tenderness to palpation over the lumbosacral spine most prominent at S1. He is unable to lift his legs against gravity; deep tendon reflexes are absent at the knees and ankles. Which of the following is the most appropriate initial step in diagnosis?

- ☐ A) Radionuclide bone scan
- ☐ B) Electromyography and nerve conduction studies
- ☐ C) CT scan of the head
- ☐ D) MRI of the lumbosacral spine
- ☐ E) Lumbar puncture



22. A 72-year-old man has had burning, aching pain in the distal extremities for 3 weeks; the pain is exacerbated by lowering the extremities and relieved by elevation. Examination shows tenderness and swelling of the fingers and wrist, knee, ankle, and toe joints; the overlying skin is warm and erythematous. There is clubbing of the fingers and toes. Which of the following is most likely to be abnormal?

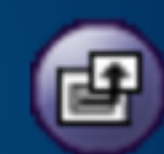
- ☐ A) Serum creatinine concentration
- ☐ B) Serum ferritin concentration
- ☐ C) Serum protein electrophoresis
- ☐ D) X-ray of the abdomen
- ☐ E) X-ray of the chest



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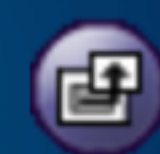
Help



Pause

23. A 62-year-old woman comes to the physician because of heartburn for 4 months and difficulty swallowing solids for 1 month. She has daily episodes of epigastric burning radiating to her chest and neck, especially at night or after ingestion of large meals. Treatment with ranitidine over the past 3 months has been ineffective. She has not had nausea, vomiting, change in bowel habits, or weight loss. She takes no other medications. She does not smoke or drink alcohol. She appears well nourished. Her temperature is 36.9°C (98.4°F), pulse is 80/min, respirations are 16/min, and blood pressure is 136/86 mm Hg. Examination shows no scleral icterus, jaundice, or lymphadenopathy. Test of the stool for occult blood is negative. Her hemoglobin concentration is 13 g/dL, serum AST activity is 18 U/L, and serum lipase activity is 224 U/L (N=14–280). Which of the following is the most appropriate next step in management?

- ☐ A) Add metoclopramide to the medication regimen
- ☐ B) Add omeprazole to the medication regimen
- ☐ C) Switch to famotidine therapy
- ☐ D) Switch to omeprazole therapy
- ☐ E) Upper endoscopy
- ☐ F) No further testing or pharmacotherapy indicated



24. A 37-year-old woman comes for a routine follow-up examination. She has renal failure secondary to polycystic kidney disease and is receiving hemodialysis. Immunization history includes diphtheria-tetanus toxoid and 23-valent pneumococcal vaccine 3 years ago and influenza virus vaccine 12 months ago. Examination shows no abnormalities. Serum hepatitis B surface antibody assay is positive. Which of the following is the most appropriate vaccine to administer at this visit?

- ☐ A) Diphtheria-tetanus toxoid
- ☐ B) Hepatitis B
- ☐ C) Influenza virus
- ☐ D) Meningococcal
- ☐ E) 23-Valent pneumococcal



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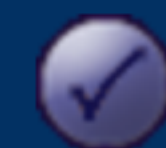
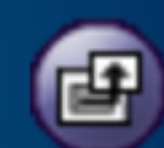
Help



Pause

25. A 67-year-old man is brought to the emergency department because of a 2-day history of episodes of vomiting blood. He has no history of similar episodes. He has esophageal adenocarcinoma, hypertension, type 2 diabetes mellitus, osteoarthritis, and coronary artery disease. He has a prosthetic, mechanical aortic valve. He is receiving radiation therapy. He notes that the therapy makes him tired and he naps several hours daily. His other medications are lisinopril, metformin, glipizide, simvastatin, warfarin, and aspirin. Which of the following is the strongest indication for anticoagulant therapy in this patient?

- ☐ A) Adenocarcinoma
- ☐ B) Coronary artery disease
- ☐ C) Decreased activity
- ☐ D) Prosthetic aortic valve
- ☐ E) Radiation therapy



26. A 22-year-old man comes to the physician because of swelling of the neck for 1 month and malaise, fever, fatigue, night sweats, and generalized pruritus for 2 weeks. Acetaminophen has provided only minimal relief of his symptoms. His temperature is 38.3°C (100.9°F), pulse is 90/min, respirations are 16/min, and blood pressure is 100/67 mm Hg. There is a 3 x 4-cm left anterior cervical lymph node. Conjunctivae are not injected, and the tympanic membranes and pharynx are not erythematous. The lungs are clear to auscultation. A grade 1/6, nonradiating systolic murmur is heard best at the upper right sternal border. Bowel sounds are slightly hyperactive. Laboratory studies show:

Hemoglobin	14 g/dL
Leukocyte count	6000/mm ³
Segmented neutrophils	60%
Bands	2%
Eosinophils	1%
Lymphocytes	33%
Monocytes	4%

Which of the following is the most likely diagnosis?

- ☐ A) Acute lymphocytic leukemia
- ☐ B) Cat-scratch disease
- ☐ C) Chronic lymphocytic leukemia
- ☐ D) Drug reaction
- ☐ E) Gastric carcinoma
- ☐ F) HIV infection
- ☐ G) Hodgkin disease
- ☐ H) Infectious mononucleosis
- ☐ I) Rubella
- ☐ J) Sarcoidosis
- ☐ K) Syphilis
- ☐ L) Systemic lupus erythematosus
- ☐ M) Tuberculosis

27. A 62-year-old man is admitted to the hospital for treatment of pneumonia. Ceftriaxone therapy is started. He also takes a thiazide diuretic for hypertension, glyburide for type 2 diabetes mellitus, and acetaminophen for osteoarthritis. Two days after admission, laboratory studies show:

Serum		
Urea nitrogen		50 mg/dL
Creatinine		2.4 mg/dL
Urine		
Na ⁺		<10 mEq/L
Fractional excretion of Na ⁺		<1
Protein		trace

Urine sediment is unremarkable. Which of the following is the most likely cause of his renal failure?

- ☐ A) Acute glomerulonephritis
- ☐ B) Acute tubular necrosis
- ☐ C) Diabetic nephropathy
- ☐ D) Interstitial nephritis
- ☐ E) Prerenal azotemia
- ☐ F) Urethral obstruction

28. A previously healthy 42-year-old woman comes to the physician 2 days after she noted a mildly tender bump on her neck. She has not had fever, chills, cough, sore throat, or palpitations. She has smoked approximately one-half pack of cigarettes daily for 20 years. She underwent a hysterectomy 2 years ago. Her only medication is conjugated estrogen. She is in no acute distress. She is 178 cm (5 ft 10 in) tall and weighs 57 kg (125 lb); BMI is 18 kg/m². Her temperature is 37°C (98.6°F), pulse is 82/min, and blood pressure is 120/78 mm Hg. Examination of the neck shows a 2-cm lesion on the left lobe of the thyroid gland. There is no cervical lymphadenopathy. Her serum thyroid-stimulating hormone concentration is 1.5 µU/mL, and serum free thyroxine concentration is 1.4 ng/dL (N=0.9–2.1). Which of the following is the most appropriate next step in management?

- ☐ A) Reexamination and repeat thyroid function tests in 1 month
- ☐ B) Measurement of thyroid radioactive iodine uptake
- ☐ C) CT scan of the neck
- ☐ D) Fine-needle aspiration of the lesion
- ☐ E) Partial thyroidectomy



29. A 47-year-old man comes to the emergency department 18 hours after the sudden onset of pain and swelling of the left knee. He has no history of recent trauma or injury to the knee. He has had two similar episodes 3 years ago and 3 months ago; both episodes resolved spontaneously. He has a 3-year history of hypertension treated with enalapril. He has smoked one pack of cigarettes daily for 20 years. He has a 10-year history of daily alcohol use. He is 173 cm (5 ft 8 in) tall and weighs 95 kg (210 lb); BMI is 32 kg/m². His temperature is 37.7°C (99.9°F), pulse is 88/min, respirations are 16/min, and blood pressure is 142/88 mm Hg. Examination shows moderate pain and swelling of the left knee; range of motion of the knee is limited by pain. Examination of the right knee shows no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Gout
- ☐ B) Osteoarthritis
- ☐ C) Psoriatic arthritis
- ☐ D) Rheumatoid arthritis
- ☐ E) Septic arthritis
- ☐ F) Systemic lupus erythematosus



30. A 47-year-old woman is brought to the emergency department 24 hours after the onset of severe abdominal pain, nausea, and vomiting. She says her pain improves somewhat with vomiting and rates her current pain as a 9 on a 10-point scale. At first, her vomitus consisted of partially digested food, but now it contains yellow fluid. She also has noticed increased abdominal girth. Her last bowel movement was 24 hours ago. She has not had fever or chills. She underwent an endodontic surgical procedure 4 days ago and has been taking acetaminophen with codeine for pain. She underwent an appendectomy at the age of 17 years and a total abdominal hysterectomy at the age of 45 years for dysfunctional uterine bleeding. She appears uncomfortable. Her temperature is 37°C (98.6°F), and respirations are 17/min. Her pulse is 120/min and blood pressure is 110/82 mm Hg while supine; her pulse is 130/min and blood pressure is 100/76 mm Hg while sitting. Examination shows dry oral mucosa. The abdomen is distended and tympanic; there is mild diffuse tenderness to palpation. Bowel sounds are high pitched. Rectal examination shows no stool in the rectal vault and no masses. Laboratory studies show:

Hematocrit	47%
Leukocyte count	13,000/mm ³
Platelet count	450,000/mm ³
Serum	
Na ⁺	145 mEq/L
K ⁺	3.4 mEq/L
Cl ⁻	103 mEq/L
HCO ₃ ⁻	35 mEq/L
Urea nitrogen	62 mg/dL
Creatinine	1.8 mg/dL

An abdominal x-ray shows distended loops of small bowel with air-fluid levels. There is no air in the rectum. Which of the following is the most likely diagnosis?

- ☐ A) Cecal volvulus
- ☐ B) Ileus
- ☐ C) Obstruction of the small bowel
- ☐ D) Pseudo-obstruction of the colon
- ☐ E) Sigmoid volvulus



31. A 58-year-old man comes to the physician because of extreme fatigue and malaise for 3 weeks. He has felt well except for a toothache 5 weeks ago treated with a root canal procedure. He has a history of a cardiac murmur first noted at the age of 19 years. His temperature is 37.8°C (100°F), pulse is 110/min, and blood pressure is 120/80 mm Hg. The lungs are clear to auscultation. Cardiac examination shows a grade 2/6, systolic ejection murmur heard best at the right second intercostal space, an S₄, and an ejection click. Laboratory studies show:

Hemoglobin	9.3 g/dL
Leukocyte count	10,000/mm ³
Segmented neutrophils	90%
Bands	10%
Erythrocyte sedimentation rate	90 mm/h
Urine blood	positive

Blood cultures are obtained. Which of the following is the most likely underlying cardiac abnormality?

- ☐ A) Calcification of a bicuspid aortic valve
- ☐ B) Calcified mitral annulus
- ☐ C) Myxomatous degeneration of the aortic valve
- ☐ D) Myxomatous degeneration of the mitral valve
- ☐ E) Myxomatous degeneration of the tricuspid valve
- ☐ F) Rheumatic disease of the mitral valve
- ☐ G) Rheumatic disease of the tricuspid valve

32. A 37-year-old man comes to the physician because of a 1-year history of progressive shortness of breath. He now has shortness of breath after walking 200 feet. He coaches a soccer team and is unable to keep up with his players on the field. He occasionally has wheezing and a nonproductive cough. He does not smoke or drink alcohol. His wife resumed smoking 18 months ago after quitting for 3 years. He has three healthy children. His father died of alcoholic cirrhosis at the age of 55 years. The patient's temperature is 36.9°C (98.5°F), pulse is 76/min, respirations are 20/min, and blood pressure is 124/76 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 90%. End-expiratory wheezes and occasional rhonchi are heard bilaterally. The remainder of the examination shows no abnormalities. An x-ray of the chest shows increased lucency and bullous changes at both lung bases. Which of the following is the most likely cause of this patient's condition?

- ☐ A) α_1 -Antitrypsin deficiency
- ☐ B) Exposure to asbestos
- ☐ C) Exposure to radon
- ☐ D) Exposure to secondhand smoke
- ☐ E) Pulmonary fibrosis



33. A 42-year-old woman is brought to the emergency department because of a 1-day history of fever, chills, headache, and joint and muscle pain. She says that she was recently hospitalized in another city for treatment of an overactive thyroid and was given two medications on discharge. She recalls no further details. Her temperature is 38°C (100.4°F), pulse is 88/min, respirations are 18/min, and blood pressure is 132/70 mm Hg. Examination shows lid lag, exophthalmos, a prominent goiter, and a supple neck. The lungs are clear to auscultation. There are no skin or joint abnormalities. Laboratory studies show:

Hematocrit	45%
Leukocyte count	1400/mm ³
Segmented neutrophils	5%
Lymphocytes	95%
Platelet count	287,000/mm ³

Which of the following is the most likely diagnosis?

- ☐ A) Autoimmune neutropenia
- ☐ B) Cyclic neutropenia
- ☐ C) Drug-induced neutropenia
- ☐ D) Radioactive iodine-induced neutropenia
- ☐ E) Thyroid storm

34. A previously healthy 32-year-old woman comes to the physician because of a 1-week history of progressive sensory loss. Initially, only her feet felt numb, but the loss of sensation has slowly ascended symmetrically to the umbilicus. She also has had urinary urgency and frequency, nocturia, and the perception of a tight band-like sensation around her midabdominal region. Examination shows slowing of left eye adduction during saccadic movement of the eyes to the right. There is diffuse hyperreflexia. Sensation to pinprick is decreased to the level of the umbilicus. The most likely explanation for the slowing of left eye adduction is a lesion at which of the following sites?

- ☐ A) Left cerebral hemisphere
- ☐ B) Left medial longitudinal fasciculus
- ☐ C) Left paramedian pontine reticular formation
- ☐ D) Right cerebellar hemisphere
- ☐ E) Right sixth cranial nerve nucleus



35. A 20-year-old woman comes to the emergency department because of a severe sore throat, hoarseness, and dysphagia for 8 hours. She appears toxic. Her temperature is 38.9°C (102°F). She has severe pharyngitis and stridor. Laboratory studies show leukocytosis. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Assay for diphtheria toxin
- ☐ B) Culture and Gram stain of sputum
- ☐ C) X-ray of the chest
- ☐ D) Barium swallow
- ☐ E) Laryngoscopy and endotracheal intubation



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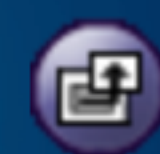
Pause

The response options for the next 2 items are the same. Select one answer for each item in the set.

For each patient with pulmonary infection, select the most likely infectious agent.

- ☐ A) Cytomegalovirus
- ☐ B) *Haemophilus influenzae*
- ☐ C) *Klebsiella pneumoniae*
- ☐ D) *Legionella pneumophila*
- ☐ E) *Mycoplasma pneumoniae*
- ☐ F) *Pneumocystis jiroveci* (formerly *P. carinii*)
- ☐ G) *Pseudomonas aeruginosa*
- ☐ H) Respiratory syncytial virus
- ☐ I) *Streptococcus pyogenes* (group A)

36. A 21-year-old woman comes to the physician because of an 8-day history of sore throat, cough, and fever. The cough is nonproductive but keeps her awake at night. She subsequently develops hoarseness. Her temperature is 38.3°C (100.9°F). Crackles are heard over the right posterior lower chest. X-rays of the chest show patchy, lower right alveolar infiltrates with increased interstitial markings. Her leukocyte count is 10,000/mm³ with 75% segmented neutrophils and 1% bands.



For each patient with pulmonary infection, select the most likely infectious agent.

- ☐ A) Cytomegalovirus
- ☐ B) *Haemophilus influenzae*
- ☐ C) *Klebsiella pneumoniae*
- ☐ D) *Legionella pneumophila*
- ☐ E) *Mycoplasma pneumoniae*
- ☐ F) *Pneumocystis jiroveci* (formerly *P. carinii*)
- ☐ G) *Pseudomonas aeruginosa*
- ☐ H) Respiratory syncytial virus
- ☐ I) *Streptococcus pyogenes* (group A)

37. In September, 5 days after returning from a business trip, a 57-year-old woman develops fever, abdominal cramps, and watery diarrhea followed by a cough 24 hours later. She has smoked one pack of cigarettes daily for 20 years. Her temperature is 39°C (102.2°F), and pulse is 90/min. Crackles are heard diffusely over both lung fields. Her leukocyte count is 16,000/mm³ with a predominance of segmented neutrophils. X-rays of the chest show alveolar infiltrates in the right upper and lower lobes and in the left lower lobe. Gram stain of sputum shows many segmented neutrophils but no predominant microorganism.



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38. A 19-year-old man comes to the physician because of a 5-day history of pain with urination and a watery urethral discharge. He has not had fever. He has no history of sexually transmitted diseases. He was sexually active with a female partner 3 weeks ago and another female partner 2 weeks ago, and he did not use condoms. He is not sure if his partners have had genitourinary symptoms. His temperature is 37.1°C (98.8°F), pulse is 80/min, and blood pressure is 122/70 mm Hg. Examination shows no lymphadenopathy. The penis and testes appear normal. There is a mucopurulent, whitish urethral discharge. Rectal examination shows a normal-sized, nontender prostate. A Gram stain of the urethral discharge shows segmented neutrophils without intracellular organisms. Which of the following is the most likely causal organism?

- ☐ A) *Chlamydia trachomatis*
- ☐ B) Herpes simplex virus 1
- ☐ C) HIV-1
- ☐ D) *Neisseria gonorrhoeae*
- ☐ E) *Treponema pallidum*



39. A 72-year-old woman is brought to the emergency department by her daughter because of a 3-day history of progressive confusion and lethargy. She underwent bilateral hip replacement 6 years ago. She takes no medications except for aspirin as needed for osteoarthritis. She is unable to answer questions. Her temperature is 36.2°C (97.2°F), pulse is 110/min and regular, respirations are 20/min, and blood pressure is 110/70 mm Hg. Examination shows a supple neck. The lungs are clear to auscultation. Cardiac examination shows no murmurs or gallops. The abdomen is soft with no guarding; bowel sounds are present but hypoactive. Deep tendon reflexes are 1+ bilaterally. Babinski sign is absent. Serum studies show:

Na ⁺	135 mEq/L
Cl ⁻	100 mEq/L
K ⁺	4.2 mEq/L
HCO ₃ ⁻	22 mEq/L
Urea nitrogen	52 mg/dL
Glucose	602 mg/dL
Creatinine	2.4 mg/dL

Which of the following is the most likely diagnosis?

- ☐ A) Alcoholic ketoacidosis
- ☐ B) Central diabetes insipidus
- ☐ C) Diabetic ketoacidosis
- ☐ D) Lactic acidosis
- ☐ E) Myxedema
- ☐ F) Nephrogenic diabetes insipidus
- ☐ G) Nonketotic hyperosmolar state
- ☐ H) Salicylate toxicity
- ☐ I) Syndrome of inappropriate secretion of ADH (vasopressin)

40. An 85-year-old woman is brought to the emergency department by paramedics 45 minutes after being found on the floor of her apartment by a home health aide. The patient says that she tripped and fell down during the night while going to the bathroom. She was on the floor for 7 hours before being found. She says that she has not been having any health problems. There is no evidence of trauma. Vital signs are within normal limits. Physical examination shows no abnormalities. Results of a complete blood count and measurement of serum electrolyte concentrations are within the reference ranges. Urinalysis shows:

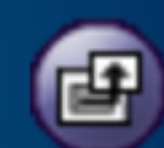
Specific gravity	1.025
Blood	2+
Protein	trace
RBC	0/hpf
WBC	0/hpf
Nitrites	negative

Which of the following is the most likely diagnosis?

- ☐ A) Glomerulonephritis
- ☐ B) Infective endocarditis
- ☐ C) Nephrolithiasis
- ☐ D) Rhabdomyolysis
- ☐ E) Urinary tract infection

41. A 72-year-old woman comes to the emergency department because of a 1-day history of fever, chills, and cough. She had pneumococcal pneumonia 1 year ago. Her temperature is 39°C (102.2°F). Examination shows bronchial breath sounds at the right lung base with increased dullness and egophony. Her leukocyte count is 87,000/mm³ (15% segmented neutrophils, 82% lymphocytes, and 3% monocytes). A Gram stain of sputum shows gram-positive, lancet-shaped diplococci. Which of the following is most likely to confirm this patient's deficit in host defenses?

- ☐ A) Assessment of segmented neutrophil function
- ☐ B) Measurement of CD4+ T-lymphocyte count
- ☐ C) Measurement of serum IgE concentration
- ☐ D) Measurement of T-lymphocyte count
- ☐ E) Quantitative immunoglobulin assay





42. A 56-year-old woman has a 1-day history of epigastric pain and vomiting. She has scleral icterus and a tender epigastrium. Laboratory studies show:

Leukocyte count	14,200/mm ³
Neutrophils	78%
Serum	
Bilirubin	
Total	4.4 mg/dL
Direct	3.3 mg/dL
Triglycerides	210 mg/dL
Amylase	1350 Somogyi U/dL (N=60–160)
Alkaline phosphatase	320 Bodansky U/dL (N=2.0–4.5)

Ultrasonography shows dilation of intrahepatic ducts. Which of the following is the most likely cause of her condition?

- ☐ A) Alcoholism
- ☐ B) Cholecystitis
- ☐ C) Choledocholithiasis
- ☐ D) Hepatoma
- ☐ E) Pancreatic carcinoma

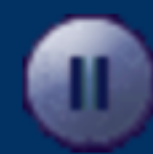
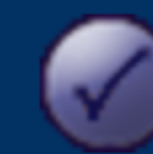
43. A 57-year-old woman with a history of alcohol abuse is brought to the emergency department by police after she was found lying in the street. On arrival, she does not respond to noxious stimuli. Her temperature is 38.1°C (100.6°F), pulse is 124/min, respirations are 20/min, and blood pressure is 94/60 mm Hg. Examination shows poor skin turgor. The oral mucosa is dry. Laboratory studies show:

Hematocrit	42%
Serum	
Urea nitrogen	58 mg/dL
Creatinine	3.4 U/L
Urine	
Color	reddish
Specific gravity	1.020
Blood	3+
Protein	1+
RBC	0–2/hpf
WBC	0–5/hpf
Hyaline casts	1/hpf

Which of the following is the most likely diagnosis?

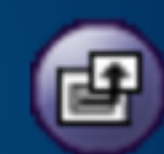
- ☐ A) Acute glomerulonephritis
- ☐ B) Acute tubular necrosis
- ☐ C) Allergic interstitial nephritis
- ☐ D) Glomerulosclerosis
- ☐ E) Pyelonephritis

44. A 67-year-old man comes to the physician as a new patient to discuss his blood pressure. He brings his medical records, which show a steady increase in systolic and diastolic blood pressure over the past 5 years. He takes no medications. He retired from his job as an insurance executive 5 years ago. He is active, exercises three times weekly, and follows a low-cholesterol diet. His blood pressure today is 160/86 mm Hg. Examination shows no other abnormalities. Which of the following is the most critical factor in the etiology of this patient's hypertension?
- ☐ A) Decreased diameter of the vascular lumen
 - ☐ B) Decreased vascular compliance
 - ☐ C) Increased deposition of lipids in arterial smooth muscle
 - ☐ D) Increased deposition of lipofuscin in cardiac myocytes
 - ☐ E) Increased intimal thickening of the arteries



45. A 24-year-old woman comes to the physician for a routine health maintenance examination. She feels well and has no history of serious illness. She has smoked one pack of cigarettes daily for 5 years and drinks two beers daily. She has a sedentary lifestyle. Her mother underwent bilateral knee replacement at the age of 55 years for severe osteoarthritis. The patient is 178 cm (5 ft 10 in) tall and weighs 101 kg (223 lb); BMI is 32 kg/m². Her pulse is 64/min, respirations are 18/min, and blood pressure is 148/92 mm Hg. Examination shows no abnormalities. She requests advice about how to prevent osteoarthritis. Which of the following is most likely to decrease this patient's risk for osteoarthritis?

- ☐ A) Abstinence from alcohol
- ☐ B) Regular aerobic exercise
- ☐ C) Weight loss
- ☐ D) Calcium supplementation
- ☐ E) Estrogen replacement therapy at the onset of menopause



46. A 42-year-old man is brought to the emergency department because of the sudden onset of worsening headache and left-sided weakness 1 hour ago. He has no history of serious illness and has not seen a physician for 10 years. He has smoked one pack of cigarettes daily for 25 years and drinks six beers daily. He is 180 cm (5 ft 11 in) tall and weighs 97 kg (215 lb); BMI is 30 kg/m². His pulse is 80/min, and blood pressure is 180/120 mm Hg. Examination shows mild-to-moderate weakness of the left lower aspect of the face. Muscle strength is decreased in the left upper and left lower extremities. Funduscopic examination shows an arteriovenous ratio of 1:3 and regions of mild focal spasm of some arterioles. Results of a complete blood count, platelet count, and coagulation studies are within the reference ranges. His serum total cholesterol concentration is 400 mg/dL. A CT scan of the head shows a 1.5 × 2-cm region of hyperintensity centered at the right putamen; there is mild surrounding lucency. Which of the following is most likely to decrease this patient's risk for a recurrent stroke of this type?

- ☐ A) Alcohol cessation
- ☐ B) Smoking cessation
- ☐ C) Weight loss
- ☐ D) Atorvastatin therapy
- ☐ E) Clopidogrel therapy
- ☐ F) Lisinopril therapy



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47. A 32-year-old nulligravid woman comes to the physician because of irregular menses for 3 years. Menses occur at 35- to 60-day intervals. Menses previously occurred at regular 28-day intervals. Her last menstrual period was 6 weeks ago. She has no history of serious illness but has been overweight most of her adult life. She is sexually active with her husband of 3 years and uses condoms consistently. She is 163 cm (5 ft 4 in) tall and weighs 77 kg (170 lb); BMI is 29 kg/m². Her blood pressure is 110/60 mm Hg. Physical examination shows moderate coarse hair over the upper lip, chin, and abdomen. Abdominal and pelvic examinations show no abnormalities. Serum studies show a thyroid-stimulating hormone concentration of 4 μ U/mL, β -hCG concentration of less than 2 mIU/mL, and prolactin concentration of 18 ng/mL. Pelvic ultrasonography shows no abnormalities. In addition to recommending weight loss, which of the following is the most appropriate pharmacotherapy?

- ☐ A) Oral contraceptive therapy
- ☐ B) Oral medroxyprogesterone therapy
- ☐ C) Oral prednisone therapy
- ☐ D) Oral thyroid therapy
- ☐ E) Depot leuprolide injection



48. A 57-year-old man is brought to the emergency department 30 minutes after a 3-minute episode of unconsciousness. The patient says that the episode began with an impending sense of feeling ill while watching television. He quickly became confused, swayed in his chair, and fell to the floor. He had no bowel or bladder incontinence during this episode. His mental status spontaneously recovered fully within 3 minutes. He has a 10-year history of hypertension well controlled with hydrochlorothiazide. He had a myocardial infarction 4 years ago. He is 183 cm (6 ft) tall and weighs 95 kg (210 lb); BMI is 29 kg/m². His pulse is 86/min, respirations are 12/min, and blood pressure is 124/86 mm Hg. Neurologic and cardiac examinations show normal findings. An ECG shows Q waves in leads II, III, and aVF. Which of the following is the most likely explanation for these findings?

- ☐ A) Hypoglycemia
- ☐ B) Paroxysmal ventricular tachycardia
- ☐ C) Pulmonary embolus
- ☐ D) Seizure
- ☐ E) Transient ischemic attack





49. A 62-year-old man has a 2-day history of generalized weakness and inability to walk without assistance 7 days after admission to the intensive care unit for treatment of fever, cough, and a left upper lobe consolidation. Treatment with ceftriaxone and azithromycin initially improved his symptoms. He has not had fever for the past 2 days. He has type 2 diabetes mellitus. His current medications are ceftriaxone, heparin, omeprazole, and insulin glargine. A peripheral catheter is in place; there is no central access. Vital signs are within normal limits. Examination of the skin shows no lesions. On pulmonary examination, rhonchi are heard in the left upper lobe. Laboratory studies show:

	Day 5	Day 7
Hemoglobin	10.8 g/dL	10 g/dL
Leukocyte count	11,500/mm ³	12,000/mm ³
Platelet count	330,000/mm ³	90,000/mm ³

Which of the following is the most appropriate next step in management?

- ☐ A) Add epoetin alfa to the medication regimen
- ☐ B) Discontinue ceftriaxone and begin physical therapy
- ☐ C) Switch from ceftriaxone to piperacillin and tazobactam therapy
- ☐ D) Switch from heparin to direct thrombin inhibitor therapy
- ☐ E) Switch from heparin to low-molecular-weight heparin therapy

50. An asymptomatic 24-year-old man who is HIV positive comes to the physician for a follow-up examination. He has been seen by a physician regularly since he was diagnosed 3 years ago and has been following all recommendations. He currently takes no medications. Examination shows no abnormalities. Laboratory studies show:

Hematocrit	42%
Leukocyte count	4000/mm ³
CD4+ T-lymphocyte count	550/mm ³ (Normal≥ 500)
Plasma HIV viral load	2000 copies/mL

Which of the following is the most appropriate recommendation for this patient?

- ☐ A) Follow-up examination in 6 months
- ☐ B) Two-drug antiretroviral therapy only
- ☐ C) Two-drug antiretroviral therapy and *Pneumocystis jiroveci* (formerly *P. carinii*) prophylaxis
- ☐ D) Three-drug antiretroviral therapy only
- ☐ E) Three-drug antiretroviral therapy and *P. jiroveci* prophylaxis